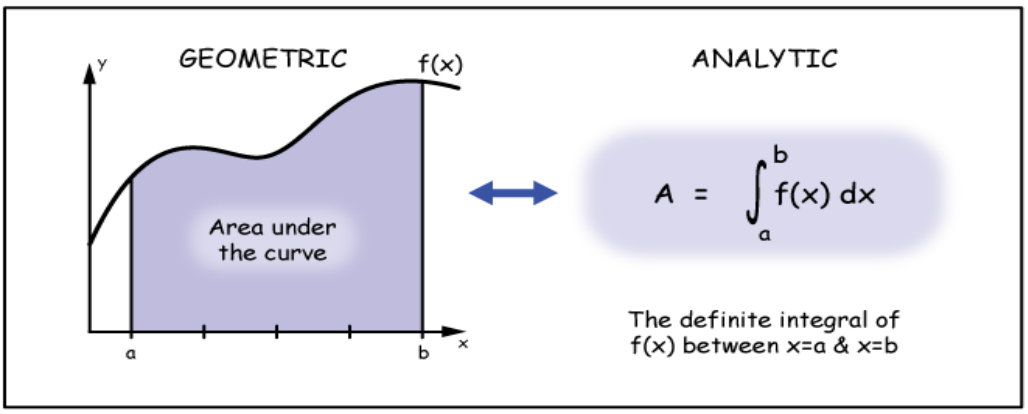
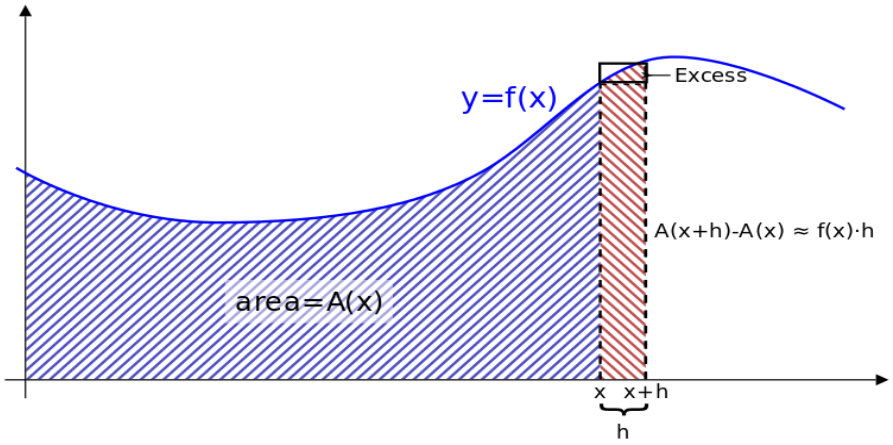
Definite Integration（定積分）

Integration is actually divided into two categories: indefinite integration (不定積分) and **definite integration (定積分)**. definite integration is commonly used for calculating areas and volumes under the curve.



How does area relate to integration? The diagram below may provide some basic ideas:



Suppose a function $A(x)$ can be used to calculate the area between the curve $y = f(x)$ and the x-axis (as shown in the figure above). Then the relationship between $f(x)$ and $A(x)$ is as follows:

$$A(x)+f(x)\cdot h \approx A(x+h) \quad \Rightarrow \quad f(x)\cdot h \approx A(x+h)-A(x)$$

When h is small enough, we have $f(x)=\lim_{h\to 0}\dfrac{A(x+h)-A(h)}{h}$

According to the first principle, $f(x)=\dfrac{d}{dx}A(x)$

Therefore, $\int f(x)\,dx = A(x)$



Not for sale

Definite Integration

Suppose $\int f(x)\,dx = F(x) + C$. Then $F(x)$ is the **primitive function (原始函數) of** $f(x)$ and its **definite integral** is evaluated as follows:

upper limit 上限

$$\int_a^b f(x)\,dx = [F(x)]_a^b$$

lower limit 下限

$$= F(b) - F(a)$$

The above definite integral is used to calculate the area between y = f(x) and the x-axis, where

- a represents the lower limit of the definite integral, i.e., the starting value of x when calculating the area;
- b represents the upper limit of the definite integral, i.e., the final value of x when calculating the area.

e.g.1 Evaluate $\int_1^2 x\,dx$.

$$\int_1^2 x\,dx = \left[\frac{x^2}{2}\right]_1^2$$

$$= \left(\frac{2^2}{2}\right) - \left(\frac{1^2}{2}\right)$$

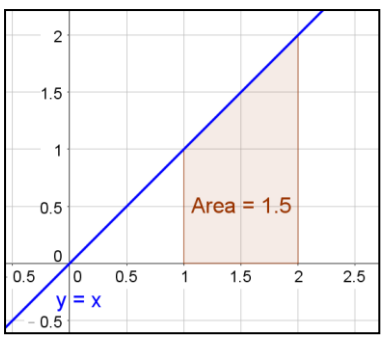
$$= 2 - \frac{1}{2}$$

$$= \frac{3}{2}$$

進行積分運算，用**中括號**將結果括起來，並在括號右方填寫上限及下限。

分別將上限及下限代入算式，並進行相減。

Note: The above answer represents the area between the line $y = x$ and the x-axis within the range from $x = 1$ to $x = 2$ (as shown in the figure on the right).



Not for sale

Definite Integration by substitution

When using integration by substitution in definite integrals, remember to **substitute both the upper limit and lower limit** simultaneously!!

e.g.2 Evaluate $\int_0^1 (2x+1)^3 dx$.

Let $u = 2x + 1$

設 u 為括號 (或根式) 裏的函數

Then $\frac{du}{dx} = 2$, i.e. $du = 2dx$

$dx = \frac{1}{2} du$

求 $\frac{du}{dx}$ 及將 dx 以 du 表示

When $x = 1$, $u = 2(1) + 1 = 3$,
when $x = 0$, $u = 2(0) + 1 = 1$

Important step:
重新計算定積分的上限及下限 !!!

$$\int_0^1 (2x+1)^3 dx = \int_1^3 (u)^3 \cdot (\frac{1}{2} du)$$

將 $2x + 1$ 換成 u, dx 換成 $\frac{1}{2} du$ 及將上下限由 **0 至 1** 換成 **1 至 3 !**

$$= \frac{1}{2} \int_1^3 u^3 \, du$$

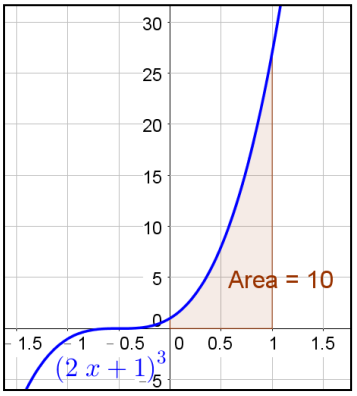
$$= \frac{1}{2} \cdot \left[\frac{u^4}{4} \right]_1^3$$

計算積分並將上限及下限代入算式然後相減

$$= \frac{1}{2} \cdot \left[\left(\frac{3^4}{4} \right) - \left(\frac{1^4}{4} \right) \right]$$

$$= 10$$

Note: The figure on the right shows the area found by the definite integral above.



Not for sale

Definite Integration by substitution

[A.Math 1983 Paper II Q.3]

Evaluate $\int_0^1 x^3\sqrt{1+3x^2}\,dx$. ***(5 marks)***

Not for sale

Definite Integration by substitution



[A.Math 1985 Paper II Q.3]

Evaluate $\int_3^4 \frac{x}{\sqrt{25-x^2}}\,dx$. ***(5 marks)***

Not for sale

Definite Integration by substitution

[A.Math 1989 Paper II Q.3]

Evaluate $\int_0^2 \frac{8x^3}{\sqrt{2x^2+1}}\,dx$. ***(5 marks)***

Not for sale

Definite Integration by trigo. sub.

Expression	Trigonometric substitution	Identity applied
$\sqrt{a^2-x^2}$	Let $x=a\sin\theta$, $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$	$1-\sin^2\theta=\cos^2\theta$
$\sqrt{a^2+x^2}$	Let $x=a\tan\theta$, $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$	$1+\tan^2\theta=\sec^2\theta$
$\sqrt{x^2-a^2}$	Let $x=a\sec\theta$, $0\le\theta<\frac{\pi}{2}$ or $\pi\le\theta<\frac{3\pi}{2}$	$\sec^2\theta-1=\tan^2\theta$

e.g.3 Find $\int_0^1 \frac{1}{\sqrt{4-x^2}}\,dx$.

Let $x=2\sin\theta$, where $-\frac{\pi}{2}\le\theta\le\frac{\pi}{2}$

then $dx=2\cos\theta\,d\theta$,

when $x=0$, $\sin\theta=0$ $\therefore$ $\theta=0$

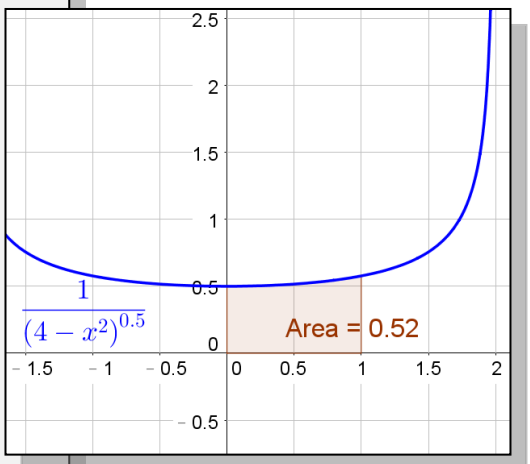
when $x=0$, $\sin\theta=\frac{1}{2}$ $\therefore$ $\theta=\frac{\pi}{6}$

設 x 為合適的三角函數

Important step:
重新計算定積分的上限及下限 !!!

$$\therefore \int_0^1 \frac{1}{\sqrt{4-x^2}}\,dx$$
$$=\int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{4-4\sin^2\theta}}\cdot 2\cos\theta\,d\theta$$
$$=\int_0^{\frac{\pi}{6}} \frac{1}{\sqrt{4\cos^2\theta}}\cdot 2\cos\theta\,d\theta$$
$$=\int_0^{\frac{\pi}{6}} \frac{1}{2\cos\theta}\cdot 2\cos\theta\,d\theta$$
$$=\int_0^{\frac{\pi}{6}} (1)\,d\theta$$
$$=[\theta]_0^{\pi/6}$$
$$=\frac{\pi}{6}-0$$
$$=\frac{\pi}{6}$$

將 x 換成 $\sin\theta$, dx 換成 $2\cos\theta d\theta$ 及更換上下限!



Not for sale

Definite Integration by trigo. sub.

[A.Math 2002 Q.4]

Evaluate $\int_0^{\frac{1}{2}} \frac{1}{\sqrt{1-x^2}}\,dx$. ***(4 marks)***

Not for sale

Definite Integral : Integration by trigo. sub.



[A.Math 1987 Paper II Q.4]

Evaluate $\int_0^{\frac{1}{2}} \frac{2x^2}{\sqrt{1-x^2}}\,dx$. ***(5 marks)***

Not for sale

Definite Integration by parts

Below are the formulas for integration by parts used in definite integrals:

$$\int_a^b f(x)\,d\,g(x) = \Big[f(x)\cdot g(x)\Big]_a^b - \int_a^b g(x)\,d\,f(x)$$

e.g.4 Find $\int_1^e x\ln x\,dx$.

Let $f(x)=\ln x$ and $d\,g(x)=x\,dx$

then $\dfrac{d}{dx}g(x)=x$

$$g(x)=\frac{x^2}{2}$$

為分部積分尋找合適的 $f(x)$ 及 $g(x)$，此時暫不需理會定積分的上下限。

Therefore

$$\int_1^e x\ln x\,dx$$

$$=\int_1^e \ln x\,d(\frac{x^2}{2})$$

$$=\left[\ln x\cdot(\frac{x^2}{2})\right]_1^e-\int_1^e(\frac{x^2}{2})\,d(\ln x)$$

使用**定積分版本**的分部積分法公式 **(緊記寫下 [$f(x)g(x)$] 的上下限 !)**

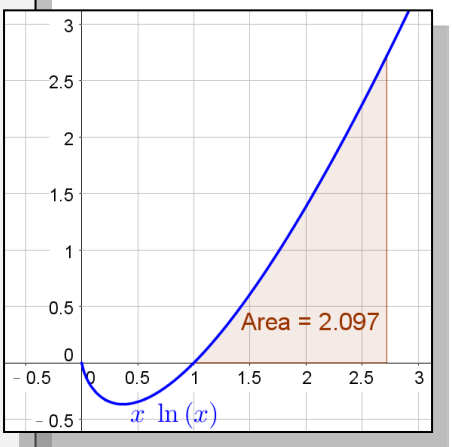
$$=\left[\frac{1}{2}x^2\ln x\right]_1^e-\int_1^e(\frac{x^2}{2})\frac{1}{x}dx$$

$$=\left(\frac{1}{2}e^2\ln e-\frac{1}{2}(1)^2\ln 1\right)-\int_1^e\frac{x}{2}dx$$

$$=\left(\frac{e^2}{2}-0\right)-\left[\frac{x^2}{4}\right]_1^e$$

$$=\frac{e^2}{2}-\left(\frac{e^2}{4}-\frac{1}{4}\right)$$

$$=\frac{e^2+1}{4}$$



Not for sale



Definite Integration by parts



Evaluate $\int_e^{e^2} \frac{\ln x}{x^2} dx$

Not for sale

More Rules of Definite Integration

Rule No. 1 :

$$\int_a^a f(x)\,dx = 0$$

***Proof*:** $\int_a^a f(x)\,dx =$

Rule No. 2 :

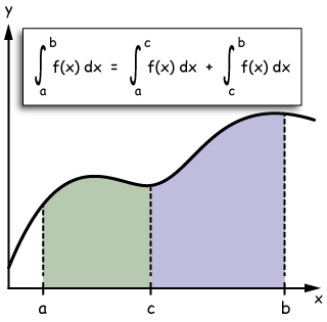
$$\int_a^b f(x)\,dx = -\int_b^a f(x)\,dx$$

***Proof*:** **L.H.S.:** $\int_a^b f(x)\,dx =$

R.H.S.: $\int_b^a f(x)\,dx =$

Rule No. 3 :

$$\int_a^c f(x)\,dx + \int_c^b f(x)\,dx = \int_a^b f(x)\,dx$$



***Proof*:** **L.H.S.:** $\int_a^c f(x)\,dx + \int_c^b f(x)\,dx$

$=$

R.H.S.: $\int_a^b f(x)\,dx =$

Not for sale

More Rules of Definite Integration

[M2 2020 Q .10(a)(b)]

(a) Using integration by substitution, prove that

$$\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \ln\left(\sin\left(\frac{\pi}{4}-x\right)\right)\mathrm{d}x = \int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \ln(\sin x)\,\mathrm{d}x \ . \qquad \textit{(3 marks)}$$

(b) Using (a), evaluate $\int_{\frac{\pi}{12}}^{\frac{\pi}{6}} \ln(\cot x - 1)\,\mathrm{d}x$. *(3 marks)*

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More Rules of Definite Integration

Definite integrals have some interesting properties, and here is one of them:

$$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$$

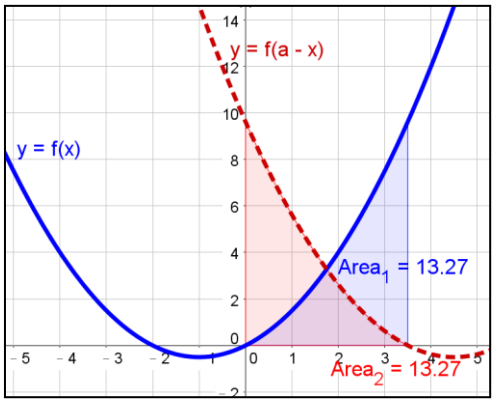
GeoGebra applet:

Proof: Let $u = a - x$, then $du = -dx$,

when $x = a$, $u = a - a = 0$;

when $x = 0$, $u = a - 0 = a$.

$$\begin{aligned}\int_0^a f(x)dx &= \int_a^0 f(a-u)(-du)\\ &= -\int_0^a f(a-u)(-du)\\ &= \int_0^a f(a-u)du\\ &= \int_0^a f(a-x)dx\end{aligned}$$

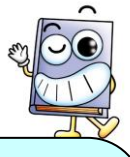


e.g.5 By considering $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$, evaluate $\int_0^{\frac{\pi}{2}} \sin^2 x\,dx$.

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^2 x\,dx &= \int_0^{\frac{\pi}{2}} \sin^2\left(\frac{\pi}{2}-x\right)dx\\ &= \int_0^{\frac{\pi}{2}} \cos^2 x\,dx\end{aligned}$$

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \sin^2 x\,dx + \int_0^{\frac{\pi}{2}} \cos^2 x\,dx &= \int_0^{\frac{\pi}{2}} (\sin^2 x + \cos^2 x)\,dx\\ &= \int_0^{\frac{\pi}{2}} 1\,dx\\ &= [x]_0^{\pi/2}\\ &= \frac{\pi}{2} - 0\\ &= \frac{\pi}{2}\end{aligned}$$

$$\therefore \int_0^{\frac{\pi}{2}} \sin^2 x\,dx = \frac{1}{2}\left(\frac{\pi}{2}\right) = \frac{\pi}{4}$$



Important:
此類題型主要使用定積分法則，而避開繁瑣的不定積分，箇中有既定的計算模式技巧，同學們須多加留意 !!!

Not for sale

More Rules of Definite Integration

[A.Math 1986 Paper II Q .8]

(a) Show that $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. ***(4 marks)***

(b) Using the result in (a), or otherwise, evaluate the following integrals:

(i) $\int_0^{\pi} \cos^{2n+1} x\,dx$, where n is a positive integer,

(ii) $\int_0^{\pi} x\sin^2 x\,dx$,

(iii) $\int_0^{\frac{\pi}{2}} \frac{\sin x}{\sin x + \cos x}\,dx$. ***(16 marks)***

Not for sale

King Property of Definite Integration

The most famous and advanced property about definite integration is called "King Property" and here it is:

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$$

Proof:

Let $u = a + b - x$,

then $du = -dx$.

When $x = a$, $\quad u = a + b - a = b$;

When $x = b$, $\quad u = a + b - b = a$.

$$\int_a^b f(x)dx = \int_b^a f(a+b-u)(-du)$$ ← Integration by substition

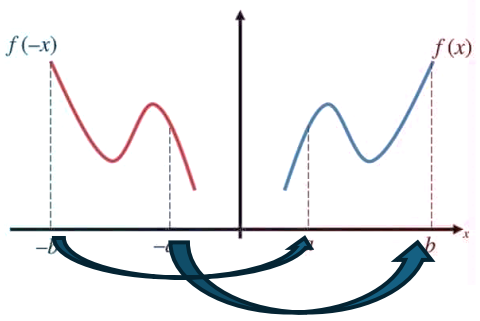
$$= -\int_a^b f(a+b-u)(-du)$$ ← Interchange the upper and lower limits

$$= \int_a^b f(a+b-u)du$$

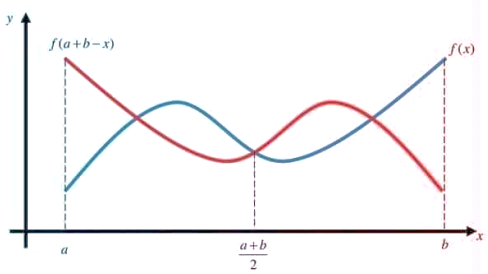
$$= \int_a^b f(a+b-x)dx$$ ← Change of dummy variable

Graphical representation:

Translate f($-x$) to the right by $(a + b)$ units:
f($-x$) ➔ f($-(x - (a + b))$) = f($a + b - x$)



$+ (a + b)$ units:

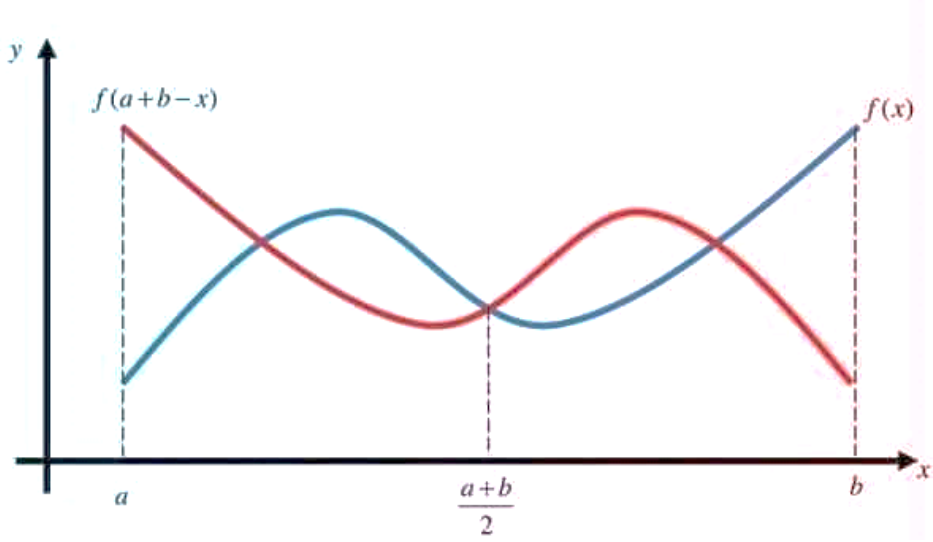


The areas under the two curves are equal:

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$$

Not for sale

King Property of Definite Integration



Note that $\int_a^{\frac{a+b}{2}} f(a+b-x)\,dx = \int_{\frac{a+b}{2}}^{b} f(x)dx$ (by considering their areas).

Therefore, we have

$$\begin{aligned}\int_a^b f(x)dx &= \int_a^{\frac{a+b}{2}} f(x)dx + \int_{\frac{a+b}{2}}^{b} f(x)dx \\ &= \int_a^{\frac{a+b}{2}} f(x)dx + \int_a^{\frac{a+b}{2}} f(a+b-x)dx \\ &= \int_a^{\frac{a+b}{2}} \left[\, f(x) \;+\; f(a+b-x) \,\right] dx\end{aligned}$$

Similarly, we have $\int_a^b f(x)dx = \int_{\frac{a+b}{2}}^{b} \left[\, f(x) \;+\; f(a+b-x) \,\right] dx$.

The above property is useful to evaluate definite integration when:

1. $f(x)$ is very difficult to be integrated directly, e.g. $\int_0^{\frac{\pi}{2}} \ln(\tan x)dx$

2. With provided lower and upper limits a and b, $f(x)+f(a+b-x)$ can be simplified, e.g.

$$\begin{aligned}\int_0^{\frac{\pi}{2}} \ln(\tan x)dx &= \int_0^{\frac{\pi}{4}} \left[\ln(\tan x) + \ln(\tan(\frac{\pi}{2}-x)) \right] dx \\ &= \int_0^{\frac{\pi}{4}} \left[\ln(\tan x) + \ln(\frac{1}{\tan x}) \right] dx \\ &= \int_0^{\frac{\pi}{4}} (0)dx = 0\end{aligned}$$

Not for sale



King Property of Definite Integration

[Modified from M2 2016 Q.10 (b) and (c)]

e.g.6 Evaluate $\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx$.

$$\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx$$

$$= \int_0^{\frac{\pi}{8}} \left[\ln(1+\tan x) + \ln(1+\tan(\frac{\pi}{4}-x)) \right] dx$$ ← King Property

$$= \int_0^{\frac{\pi}{8}} \left[\ln(1+\tan x) + \ln(1+\frac{\tan\frac{\pi}{4}-\tan x)}{1+\tan\frac{\pi}{4}\tan x}) \right] dx$$ ← Compound Angle Formula

$$= \int_0^{\frac{\pi}{8}} \left[\ln(1+\tan x) + \ln(1+\frac{1-\tan x)}{1+\tan x}) \right] dx$$ ← $\tan\frac{\pi}{4}=1$

$$= \int_0^{\frac{\pi}{8}} \left[\ln(1+\tan x) + \ln(\frac{1+\tan x+1-\tan x)}{1+\tan x}) \right] dx$$

$$= \int_0^{\frac{\pi}{8}} \left[\ln(1+\tan x) + \ln(\frac{2}{1+\tan x}) \right] dx$$

$$= \int_0^{\frac{\pi}{8}} \left[\ln\left((1+\tan x)(\frac{2}{1+\tan x}) \right) \right] dx$$ ← $\ln M + \ln N = \ln MN$

$$= \int_0^{\frac{\pi}{8}} \ln 2 dx$$

$$= \left[x\ln 2 \right]_0^{\frac{\pi}{8}}$$

$$= \frac{\pi \ln 2}{8}$$

Note:

You may also show that

$$\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx = \int_0^{\frac{\pi}{4}} \ln\left(1+\tan\left(\frac{\pi}{4}-x\right)\right)dx = \int_0^{\frac{\pi}{4}} \ln\left(\frac{2}{1+\tan x}\right)dx \text{ and hence}$$

$$\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx = \frac{1}{2}\int_0^{\frac{\pi}{4}} \left[\ln(1+\tan x) + \ln(\frac{2}{1+\tan x}) \right] dx$$

$$= \frac{\ln 2}{2}\int_0^{\frac{\pi}{4}} dx$$

$$= \frac{\pi \ln 2}{8}$$

King Property of Definite Integration



[M2 2022 Q.10(b)(c)]

Evaluate $\int_0^{\pi} \cos^2 x \cos 2x \, dx$.

Hence, evaluate $\int_0^{\pi} x \cos^2 x \cos 2x \, dx$. ***(6 marks)***

HKCEE Questions (Level 1)

1. ***[A.Math 1988 Paper II Q.6]***

Evaluate $\int_0^2 \frac{x^2}{\sqrt{9-x^3}}\,dx$. ***(6 marks)***

Not for sale

HKCEE Questions (Level 2)

2. ***[A.Math 1981 Paper II Q.3]***

Evaluate $\int_0^9 \frac{x}{\sqrt{9-x}}\,dx$. ***(6 marks)***

Not for sale

HKDSE Questions (level 2)

3. ***[M2 2015 Q.3]***

(a) Find $\int \frac{1}{e^{2u}} du$.

(b) Using integration by substitution, evaluate $\int_1^9 \frac{1}{\sqrt{x}e^{2\sqrt{x}}} dx$. ***(7 marks)***

HKCEE Questions (Level 2)

4. ***[A.Math 1991 Paper II Q.2]***

Evaluate $\int_0^{\frac{\pi}{2}} (\sin x + \cos x)^2 \, dx$. ***(5 marks)***

Not for sale

HKCEE Questions (level 2)

5. ***[A.Math 1998 Paper II Q.6]***

Evaluate $\int_0^{\frac{\pi}{2}} \cos^5\theta \sin^2\theta \, d\theta$. ***(6 marks)***

Not for sale

HKCEE Questions (Level 2)

6. ***[A.Math 1982 Paper II Q.6]***

Given that $I = \int_0^{\frac{\pi}{2}} \frac{\sin^3 x}{\sin x + \cos x}\, dx = \int_0^{\frac{\pi}{2}} \frac{\cos^3 x}{\sin x + \cos x}\, dx$. By considering the sum of the above two integrals, evaluate the value of *I*. ***(6 marks)***

HKCEE Questions (Level 3)

7. ***[A.Math 2008 Q .17]***

(a) Let $f(x) = x\sin x$.

(i) Show that $\int_0^{\pi} [f(x) + f(\pi - x)]\, dx = 2\pi$.

(ii)

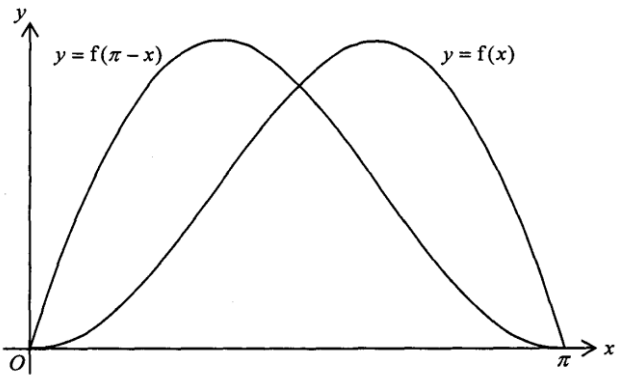


Figure 6

Figure 6 shows the graphs of $y = f(x)$ and $y = f(\pi - x)$ for $0 \le x \le \pi$.
Using (a)(i), and by considering the symmetry of the graphs of $y = f(x)$ and $y = f(\pi - x)$, write down the value of $\int_0^{\pi} x\sin x\, dx$. ***(5 marks)***

(b) Find $\frac{d}{dx}(x^2 \sin x)$, and hence evaluate $\int_0^{\pi} x^2 \cos x\, dx$. ***(4 marks)***

HKCEE Questions (level 3)

Not for sale

HKDSE Questions (Level 3)

8. ***[M2 2016 Q .10]***

(a) Let $f(x)$ be a continuous function defined on the intervals $[0, a]$, where a is a positive constant. Prove that $\int_0^a f(x)dx = \int_0^a f(a-x)dx$. ***(3 marks)***

(b) Prove that $\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx = \int_0^{\frac{\pi}{4}} \ln(\frac{2}{1+\tan x})dx$. ***(3 marks)***

(c) Using (b), prove that $\int_0^{\frac{\pi}{4}} \ln(1+\tan x)dx = \frac{\pi \ln 2}{8}$. ***(3 marks)***

(d) Using integration by parts, evaluate $\int_0^{\frac{\pi}{4}} \frac{x\sec^2 x}{1+\tan x}dx$. ***(3 marks)***

HKDSE Questions (level 3)

Not for sale

HKDSE Questions (Level 3)

9. ***[M2 2021 Q .9]***

(a) Let $\frac{-\pi}{2} < \theta < \frac{\pi}{2}$.

(i) Find $\frac{\mathrm{d}}{\mathrm{d}\theta}\ln(\sec\theta + \tan\theta)$.

(ii) Using the result of (a)(i), find $\int \sec\theta \,\mathrm{d}\theta$. Hence, find $\int \sec^3\theta \,\mathrm{d}\theta$.

(4 marks)

(b) Let $g(x)$ and $h(x)$ be continuous functions defined on $\mathbf{R}$ such that $g(x) + g(-x) = 1$ and $h(x) = h(-x)$ for all $x \in \mathbf{R}$.

Using integration by substitution, prove that $\int_{-a}^{a} g(x)h(x)\,\mathrm{d}x = \int_{0}^{a} h(x)\,\mathrm{d}x$ for any $a \in \mathbf{R}$.

(3 marks)

(c) Evaluate $\int_{-1}^{1} \frac{3^x x^2}{(3^x + 3^{-x})\sqrt{x^2+1}}\,\mathrm{d}x$. (5 marks)

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HKCEE Questions (Level 4)

10. ***[A.Math 1985 Paper II Q .8]***

(a) Evaluate $\int_0^{\frac{\pi}{2}} \sin^3 t \cos^4 t \, dt$. ***(4 marks)***

(b) Using the substitution $t = \frac{\pi}{2} - u$, show that

$$\int_0^{\frac{\pi}{2}} \cos^3 t \sin^4 t \, dt = \int_0^{\frac{\pi}{2}} \sin^3 t \cos^4 t \, dt .$$ ***(4 marks)***

(c) Show that $\int_{-\frac{\pi}{2}}^{0} \cos^3 t \sin^4 t \, dt = \int_0^{\frac{\pi}{2}} \cos^3 t \sin^4 t \, dt$

and $\int_{-\frac{\pi}{2}}^{0} sin^3 t \cos^4 t \, dt = -\int_0^{\frac{\pi}{2}} \sin^3 t \cos^4 t \, dt$. ***(6 marks)***

(d) Using the above result, or otherwise, evaluate

$$\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{8} sin^3 2t(\sin t + \cos t) \, dt .$$ ***(6 marks)***

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11. ***[A.Math 1996 Paper II Q .9]***

(a) Evaluate $\int_0^{\pi} \sin^5 x \, dx$. ***(4 marks)***

(b) Using the substitution $u = \pi - x$ and the result of (a), evaluate

$\int_0^{\pi} x \sin^5 x \, dx$. ***(5 marks)***

(c) By differentiating $y = x^2 \sin^5 x$ with respect to x and using the result of (b),

evaluate $\int_0^{\pi} x^2 \sin^4 x \cos x \, dx$. ***(5 marks)***

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12. ***[A.Math 1990 Paper II Q .9]***

(a) (i) Evaluate $\int_0^{\pi} \cos^2 x \, dx$.

(ii) Using the substitution $u = \pi - x$, evaluate $\int_0^{\pi} x\cos^2 x \, dx$.

(7 marks)

(b) Show that

(i) $\int_{\pi}^{2\pi} x\cos^2 x \, dx = \pi\int_0^{\pi} \cos^2 x \, dx + \int_0^{\pi} x\cos^2 x \, dx$.

(ii) $\int_0^{2\pi} x\cos^2 x \, dx = \pi^2$. ***(6 marks)***

(c) Using the result of (b)(ii), evaluate $\int_0^{\sqrt{2\pi}} x^3 \cos^2 x^2 \, dx$. ***(3 marks)***

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13. ***[A.Math 1980 Paper II Q .8]***

(a) Suppose $f(x) = f(a-x)$ for all real values of x. By using the substitution $u = a - x$, show that $\int_0^a xf(x)\,dx = a\int_0^a f(u)\,du - \int_0^a uf(u)\,du$.

Hence deduce that $\int_0^a xf(x)\,dx = \frac{a}{2}\int_0^a f(x)\,dx$ ***(6 marks)***

(b) By using the substitution $u = x - \frac{\pi}{2}$, show that

$$\int_{\frac{\pi}{2}}^{\pi} \frac{\sin^4 x}{\sin^4 x + \cos^4 x}\,dx = \int_0^{\frac{\pi}{2}} \frac{\cos^4 u}{\sin^4 u + \cos^4 u}\,du.$$

By using the above result and consider

$$\int_a^b f(x)\,dx = \int_a^c f(x)\,dx + \int_c^b f(x)\,dx,$$

evaluate $\int_0^{\pi} \frac{\sin^4 x}{\sin^4 x + \cos^4 x}\,dx$. ***(9 marks)***

(c) Using (a) and (b), evaluate $\int_0^{\pi} \frac{x\sin^4 x}{\sin^4 x + \cos^4 x}\,dx$. ***(5 marks)***

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14. ***[A.Math 1998 Paper II Q .9]***

(a) Let a be a positive number.

(i) Show that $\int_{-a}^{0} f(x)\,dx = \int_{0}^{a} f(-x)\,dx$.

(ii) If $f(x) = f(-x)$ for $-a \le x \le a$, show that $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$.

(5 marks)

(b) Using the substitution $t = \frac{\sqrt{3}}{3}\tan\theta$, show that

$$\int_{0}^{1} \frac{dt}{1+3t^2} = \frac{\sqrt{3}\pi}{9}.$$ ***(3 marks)***

(c) Given $I_1 = \int_{0}^{1} \frac{1-t^2}{1+3t^2}\,dt$ and $I_2 = \int_{0}^{1} \frac{t^2}{1+3t^2}\,dt$.

(i) Without evaluating I_1 and I_2,

(1) show that $I_1 + 4I_2 = 1$;

(2) using the result of (b), evaluate $I_1 + I_2$.

(ii) Using the result of (c)(i), or otherwise, evaluate I_2. ***(4 marks)***

(d) Evaluate $\int_{-1}^{1} \frac{1+t^2}{1+3t^2}\,dt$. ***(4 marks)***

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HKDSE Questions (Level 4)

15. *[M2 2023 Q .12]*

(a) Let a be a non-zero constant. Prove that $\int_0^1 x^2 e^{ax}\,dx = \frac{(a^2-2a+2)e^a-2}{a^3}$. (3 marks)

(b) Using (a) and integration by substitution, evaluate $\int_0^{e-1} x\left(\ln(1+x)\right)^2\,dx$. (4 marks)

(c) Evaluate $\int_0^{\frac{\pi}{2}} \left(\ln(1+(e-1)\cos x)\right)^2 \sin 2x\,dx$. (3 marks)

(d) Evaluate $\int_{\frac{\pi}{2}}^{\pi} \left(\ln(1+(e-1)\sin x)\right)^2 \sin 2x\,dx$. (3 marks)

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